

PORT-HAMILTONIAN DISSIPATIVE LANDING WITH CONFORMAL METRIC BARRIERS AND RANDOMIZED NYSTRÖM PROJECTIONS

Anonymous authors

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ABSTRACT

Sampling probability distributions restricted to non-linear Riemannian submanifolds with equality and inequality constraints is central to Bayesian inference, molecular dynamics, and constrained machine learning. Existing projection-free Langevin samplers suffer from severe numerical stiffness under high-gain proportional landing drifts, active-set chattering and offset (ϵ) bias near inequality boundaries, $O(m^3)$ matrix inversion bottlenecks, and numerical breakdown when the Linear Independence Constraint Qualification (LICQ) degrades. We propose **PH-CNR-OLLA**, a projection-free manifold sampler that resolves these failure modes with four ingredients: (i) a *Port-Hamiltonian* state-space structure that converts stiff normal energy into damped rotational momentum, enabling stable execution at large step sizes; (ii) a C^∞ -smooth *Conformal Riemannian metric* that warps kinetic space near boundaries so that infeasible regions are naturally unreachable, removing active-set switching and ϵ -bias; (iii) a randomized *Nyström* low-rank tangent projection with rank-truncated SVD, reducing projection cost to $O(r^3 + rmd)$ while remaining stable under LICQ degradation; and (iv) Rademacher-Hutchinson curvature estimation with lazy Jacobian refreshing for scaling to $d > 100,000$. Across 2D manifolds, high-dimensional stress tests ($d \leq 700$), polymer systems ($d \leq 90$), and fairness-constrained Bayesian networks ($d \leq 100,002$), PH-CNR-OLLA attains state-of-the-art sampling accuracy, stays stable at step sizes where baseline landing samplers diverge, reduces constraint violations by up to $5\times$, eliminates boundary chattering, and yields a $\sim 1.9\times$ speedup.

1 INTRODUCTION

Sampling from complex probability distributions restricted to non-linear Riemannian submanifolds embedded in Euclidean space $\mathbb{R}^d$ is a fundamental problem in modern computational science, Bayesian statistics, molecular dynamics, and constrained machine learning Girolami & Calderhead (2011); Diaconis et al. (2012). In these applications, the target density is supported on a compact, connected Riemannian submanifold defined by a set of smooth equality constraints $h(x) = 0 \in \mathbb{R}^m$ and inequality constraints $g(x) \leq 0 \in \mathbb{R}^l$:

$$\Sigma := \{x \in \mathbb{R}^d \mid h(x) = 0 \in \mathbb{R}^m, g(x) \leq 0 \in \mathbb{R}^l\}. \quad (1)$$

The target density takes the form $\rho_\Sigma(x) \propto \exp(-f(x)) d\sigma_\Sigma$, where $f: \mathbb{R}^d \rightarrow \mathbb{R}$ is a potential function and $d\sigma_\Sigma$ denotes the induced Riemannian volume measure on Σ . Efficient sampling from $\rho_\Sigma(x)$ is essential for diverse downstream tasks, ranging from geometric molecular conformer generation Xu et al. (2022) to individual fairness enforcement in Bayesian deep neural networks Doherty et al. (2023).

Existing sampling approaches primarily fall into two methodological paradigms: projection-based Markov Chain Monte Carlo (MCMC) algorithms Zappa et al. (2017) and interior-point barrier or mirror-map samplers Hsieh & Cevher (2018); Ahn & Chewi (2020). Projection-based methods (e.g., Constrained Langevin and Constrained Hamiltonian Monte Carlo) map unconstrained proposal steps back onto Σ via iterative Newton-Raphson projections. While mathematically rigorous, these pro-

jections incur an expensive $O(N_{\text{newton}}m^3)$ computational overhead at every step and frequently suffer solver step-size collapses when encountering high ambient dimensions or non-linear constraint boundaries. Conversely, barrier and mirror-map approaches reformulate constraints via logarithmic penalties or domain mappings; however, they often distort target distributions near boundaries by introducing artificial repulsive potentials or inflate ambient space via slack variable additions.

To circumvent iterative projections without domain transformations, recent projection-free frameworks such as Overdamped Langevin with LAnding (OLLA) Jeon et al. (2025) and orthogonal-space variational gradient descent Zhang et al. (2022) introduce continuous-time Stochastic Differential Equations (SDEs) equipped with deterministic landing drifts $-\alpha \nabla h(x)^T G(x)^{-1} h(x)$, where $G(x) = \nabla h(x) \nabla h(x)^T$ is the constraint Gram matrix and $\alpha > 0$ is a landing rate. These dynamics pull off-manifold trajectories back toward Σ exponentially without explicit projection solvers.

Despite avoiding explicit Newton-Raphson solves, current projection-free landing samplers suffer from four major methodological bottlenecks:

1. **Numerical Stiffness and Discretization Bias:** High-gain proportional landing forces ($\alpha \gg 1$) introduce severe numerical stiffness into the continuous SDE. In discrete time, this requires minuscule integration step sizes $\eta \ll 1/\alpha$ to avoid instability; under moderate step sizes, proportional landing drift generates significant off-manifold discretization bias.
2. **Boundary Chattering and Hyperparameter Bias (ϵ):** Inequality constraints $g(x) \leq 0$ are typically enforced via active-set indicator logic I_x and an offset parameter $\epsilon > 0$. This discrete active-set switching introduces jump discontinuities into the drift vector field, causing severe boundary chattering and hyperparameter-induced boundary bias.
3. **Heavy $O(m^3)$ Matrix Inversions and Hessian Operations:** Forming and inverting the exact Gram matrix $G(x) \in \mathbb{R}^{m \times m}$ requires $O(m^3)$ operations per step, while computing Ito-Stratonovich mean-curvature corrections requires evaluating $O(d \cdot \text{grad-cost})$ constraint Hessian trace terms.
4. **Instability Under LICQ Degradation:** When the Linear Independence Constraint Qualification (LICQ) Bergmann & Herzog (2018) degrades—for instance, near constraint singularities where constraint gradients become nearly linearly dependent—the Gram matrix $G(x)$ becomes ill-conditioned ($\text{cond}(G) \gg 10^{12}$), causing exact matrix inversions to blow up and trajectories to diverge.

To address these limitations simultaneously, we propose **PH-CNR-OLLA**, whose four technical advances map directly onto the four bottlenecks above (Figure 1). **(1) Port-Hamiltonian skew-symmetric landing** couples off-manifold error $h(x)$ to a normal momentum $p \in \mathbb{R}^m$ through a skew-symmetric $\mathcal{J}(x) = -\mathcal{J}(x)^T$ and dissipates it at rate γ , so $h(X_t) \rightarrow 0$ without gradient-descent stiffness. **(2) A conformal Riemannian metric barrier** $M(x) = I_d + \sum_{j=1}^l \psi(g_j(x)) \nabla g_j(x) \nabla g_j(x)^T$ places infeasible regions at infinite Riemannian distance, eliminating active-set switching, drift discontinuities, chattering, and ϵ -offset bias. **(3) A randomized Nyström low-rank projection** replaces the $O(m^3)$ Gram inverse with a rank- r ($r \ll m$) sketch whose truncated SVD spectrally filters vanishing singular values under LICQ degradation. **(4) Rademacher-Hutchinson curvature estimation** with lazy Jacobian refreshing evaluates the mean-curvature drift $H(x) = \text{div}_\Sigma(\Pi_r(x))$ at a cost independent of d . Across 2D geometries, high-dimensional stress tests ($d \leq 700$), a polymer system ($d \leq 90$), and fairness-constrained Bayesian logistic regression ($d \leq 100,002$), PH-CNR-OLLA eliminates chattering, stays stable under severe LICQ degradation, lowers CPU/ESS, and matches or exceeds OLLA/OLLA-H (Jeon et al., 2025), CLangevin (Zappa et al., 2017), CHMC (Zappa et al., 2017), and CGHMC on both accuracy and feasibility.

2 RELATED WORK

Sampling on constrained Riemannian manifolds has been investigated through three primary methodological paradigms: projection-based MCMC samplers, barrier/soft-penalty formulations, and projection-free landing dynamics.

2.1 PROJECTION-BASED MANIFOLD SAMPLERS

Classical Markov Chain Monte Carlo (MCMC) algorithms on Riemannian manifolds, such as Riemann Manifold Langevin and Hamiltonian Monte Carlo (RM-HMC) Girolami & Calderhead (2011); Diaconis et al. (2012); Wang et al. (2020), exploit parameter-space metric tensors to construct proposals tailored to local manifold geometry. When the target density is restricted to sub-manifolds defined by explicit non-linear constraint equations $h(x) = 0$, projection-based samplers such as Constrained Hamiltonian Monte Carlo (CHMC) Zappa et al. (2017), Constrained Langevin Dynamics (CLangevin) Zappa et al. (2017); Zhang et al. (2022), and Constrained Generalized HMC (CGHMC) enforce feasibility by projecting proposed Euclidean steps back onto the constraint surface using iterative Newton-Raphson solvers.

However, these methods require solving an $m \times m$ linear system $\nabla h(x) \nabla h(x)^T \Delta \lambda = h(x)$ at each Newton iteration, incurring an $O(N_{\text{newton}} m^3)$ computational cost. Furthermore, in high dimensions or near sharp manifold boundaries, Newton-Raphson iterations frequently fail to converge, leading to high proposal rejection rates or solver step-size collapses. In contrast, PH-CNR-OLLA completely eliminates iterative Newton-Raphson projections by converting normal constraint potential energy into momentum via a Port-Hamiltonian dissipative state-space structure.

2.2 BARRIER AND SOFT-PENALTY METHODS

To avoid exact geometric projections, interior-point barrier methods and self-concordant barrier HMC Noble et al. (2022); Gu et al. (2024); Srinivasan et al. (2024) construct barrier metrics or log-potentials that repel stochastic trajectories away from constraint boundaries. Similarly, Mirrored Langevin Dynamics Hsieh & Cevher (2018) and Mirror-Langevin algorithms Ahn & Chewi (2020) utilize mirror maps to transform constrained domains (such as simplices or compact convex bodies) into unconstrained dual spaces. In addition, variational mean-field particle frameworks Kook et al. (2024) analyze stationary distributions under interacting particle approximations and non-convex boundaries.

Despite their theoretical guarantees, standard interior log-barriers distort the target near boundaries via artificial repulsive potentials, while slack-variable reformulations inflate the ambient dimension ($d \rightarrow d + l$) and introduce stiff dynamics. PH-CNR-OLLA instead uses a C^∞ conformal metric that renders boundaries unreachable by Brownian motion without corrupting the interior density.

2.3 PROJECTION-FREE AND LANDING-BASED LANGEVIN DYNAMICS

To achieve projection-free sampling without domain transformations, recent works have explored overdamped Langevin landing dynamics Jeon et al. (2025), orthogonal-space variational gradient descent Zhang et al. (2022), and distributed stochastic projection-free optimization Jiang et al. (2022). In particular, Overdamped Langevin with LAnding (OLLA) Jeon et al. (2025) constructs continuous-time SDEs with deterministic normal landing drifts $-\alpha \nabla h(x)^T G(x)^{-1} h(x)$ to enforce equality constraints, while handling active inequality constraints $g_j(x) \geq 0$ via discrete active-set logic I_x and boundary repulsion rates $\epsilon > 0$.

However, when LICQ degrades or constraint Jacobians lose full rank (Bergmann & Herzog, 2018), the Gram matrix $G(x)$ becomes ill-conditioned and standard landing samplers break down; moreover, OLLA’s proportional landing drift is stiff at large step sizes and its discrete active-set switching induces drift discontinuities and chattering. PH-CNR-OLLA addresses these issues jointly through Port-Hamiltonian momentum channels, conformal metric barriers, and rank-truncated Nyström sketching.

3 METHODOLOGY

Figure 1 summarizes the four components of PH-CNR-OLLA, which we detail in turn below after fixing notation.

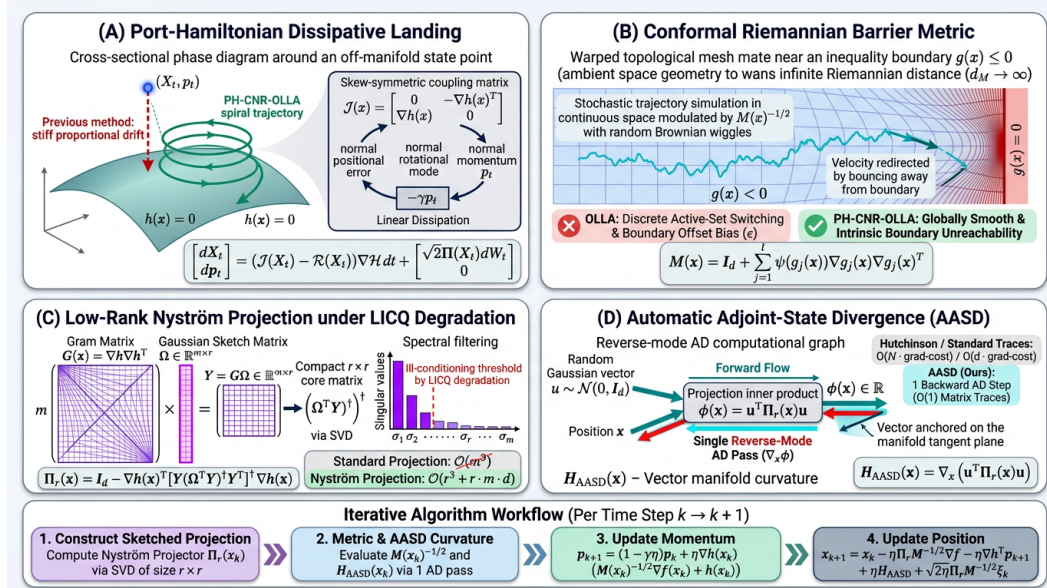


Figure 1: Overview of the four components of PH-CNR-OLLA. (A) Port-Hamiltonian dissipative landing rotates off-manifold positional error $h(x)$ into a normal-momentum channel p that is linearly damped at rate γ , replacing the stiff proportional drift of prior landing samplers with a smooth spiral onto $\{h(x) = 0\}$. (B) A C^∞ conformal Riemannian metric $M(x)$ warps ambient space so that inequality boundaries $\{g(x) = 0\}$ lie at infinite Riemannian distance, deflecting Brownian trajectories without active-set switching or ϵ -offset bias. (C) Randomized Nyström projection replaces the $O(m^3)$ Gram inverse by a sketched $r \times r$ SVD that spectrally filters vanishing singular values under LICQ degradation. (D) The mean-curvature drift is obtained via low-variance trace estimation, scaling independently of the ambient dimension.

3.1 GEOMETRIC SETUP AND PRELIMINARIES

We consider the problem of sampling from a target probability distribution supported on a compact, connected Riemannian submanifold $\Sigma \subset \mathbb{R}^d$ defined by m smooth equality constraints $h(x) = [h_1(x), \dots, h_m(x)]^T = 0 \in \mathbb{R}^m$ and l smooth inequality constraints $g(x) = [g_1(x), \dots, g_l(x)]^T \leq 0 \in \mathbb{R}^l$:

$$\Sigma := \{x \in \mathbb{R}^d \mid h(x) = 0 \in \mathbb{R}^m, g(x) \leq 0 \in \mathbb{R}^l\}. \quad (2)$$

To ensure that the geometric structures associated with Σ are well-defined, we assume that the constraints satisfy the Linear Independence Constraint Qualification (LICQ) Bergmann & Herzog (2018). Under LICQ, the constraint Jacobians $\nabla h(x) \in \mathbb{R}^{m \times d}$ and active inequality Jacobians $\nabla g_{I_x}(x) \in \mathbb{R}^{|I_x| \times d}$ have full row rank for all $x \in \Sigma$, where $I_x := \{j \in [l] \mid g_j(x) = 0\}$ denotes the active inequality index set.

At any point $x \in \Sigma$, the tangent space is $T_x \Sigma = \{v \in \mathbb{R}^d \mid \nabla h(x)v = 0, \nabla g_j(x)v = 0, \forall j \in I_x\}$. The exact orthogonal projection operator onto the tangent space of the equality surface is defined as:

$$\Pi(x) = I_d - \nabla h(x)^T G(x)^{-1} \nabla h(x), \quad \text{where } G(x) = \nabla h(x) \nabla h(x)^T \in \mathbb{R}^{m \times m} \quad (3)$$

is the equality constraint Gram matrix. The target density on Σ is expressed as $\rho_\Sigma(x) \propto \exp(-f(x)) d\sigma_\Sigma(x)$, where $f: \mathbb{R}^d \rightarrow \mathbb{R}$ is a smooth potential energy function and $d\sigma_\Sigma$ is the induced Riemannian Hausdorff volume measure. The continuous-time Langevin diffusion on Σ requires an intrinsic drift correction $H(x) = \text{div}_\Sigma(\Pi(x))$ representing the manifold mean-curvature vector field Jeon et al. (2025):

$$H(x) = -\nabla h(x)^T G(x)^{-1} [\text{Tr}(\Pi(x) \nabla^2 h_i(x))]_{i=1}^m \in \mathbb{R}^d, \quad (4)$$

which preserves geometric Itô-Stratonovich consistency during stochastic integration.

3.2 PORT-HAMILTONIAN DISSIPATIVE LANDING DYNAMICS

Standard landing-based samplers enforce equality constraints via proportional high-gain landing drifts $-\alpha \nabla h(x)^T G(x)^{-1} h(x)$ with landing rate $\alpha \gg 1$. This proportional control loop induces

high numerical stiffness in discrete time, forcing the step size to satisfy $\eta < 2/\alpha$ to avoid explosive divergence.

To resolve this stiffness, we embed the off-manifold dynamics into a Port-Hamiltonian (PH) state-space structure. We augment the position state $x \in \mathbb{R}^d$ with an auxiliary normal momentum vector $p \in \mathbb{R}^m$ associated with the equality constraints $h(x) = 0$. We define the extended system Hamiltonian $\mathcal{H}(x, p)$ as:

$$\mathcal{H}(x, p) = f(x) + \frac{1}{2}\|h(x)\|_2^2 + \frac{1}{2}\|p\|_2^2. \quad (5)$$

We then construct the continuous-time Port-Hamiltonian Stochastic Differential Equation (SDE) governed by skew-symmetric inter-domain coupling $\mathcal{J}(x)$ and block dissipation $\mathcal{R}(x)$:

$$\begin{bmatrix} dX_t \\ dp_t \end{bmatrix} = (\mathcal{J}(X_t) - \mathcal{R}(X_t)) \begin{bmatrix} \nabla_x \mathcal{H}(X_t, p_t) \\ \nabla_p \mathcal{H}(X_t, p_t) \end{bmatrix} dt + \begin{bmatrix} \sqrt{2}\Pi(X_t)dW_t \\ 0 \end{bmatrix}, \quad (6)$$

where W_t is standard d -dimensional Brownian motion, and the structure matrices are:

$$\mathcal{J}(x) = \begin{bmatrix} 0 & -\nabla h(x)^T \\ \nabla h(x) & 0 \end{bmatrix} = -\mathcal{J}(x)^T, \quad \mathcal{R}(x) = \begin{bmatrix} \Pi(x) & 0 \\ 0 & \gamma I_m \end{bmatrix} \succeq 0, \quad (7)$$

with momentum damping coefficient $\gamma > 0$. Unrolling the matrix SDE system (6) yields explicit coupled vector equations:

$$\begin{aligned} dX_t &= -(\Pi(X_t)\nabla f(X_t) + \nabla h(X_t)^T p_t) dt + H(X_t)dt + \sqrt{2}\Pi(X_t)dW_t, \\ dp_t &= (\nabla h(X_t)(\nabla f(X_t) + h(X_t)) - \gamma p_t) dt. \end{aligned} \quad (8)$$

Energy Dissipation and Non-Stiff Landing Property. The skew-symmetric block $\mathcal{J}(x)$ acts as an energy-conserving rotational channel: off-manifold positional energy $\frac{1}{2}\|h(X_t)\|_2^2$ is smoothly transformed into kinetic energy $\frac{1}{2}\|p_t\|_2^2$ without high-gain gradient forces. The linear damping term $-\gamma p_t$ subsequently drains this kinetic energy exponentially.

Proposition 1 (Energy Dissipation of Normal Momentum). *In the absence of stochastic tangential noise and intrinsic potential gradients ($\nabla f = 0$), the time derivative of the positional constraint energy under the Port-Hamiltonian flow satisfies:*

$$\frac{d}{dt} \left(\frac{1}{2}\|h(X_t)\|_2^2 + \frac{1}{2}\|p_t\|_2^2 \right) = -\gamma\|p_t\|_2^2 \leq 0. \quad (9)$$

Thus, off-manifold positional errors $h(X_t)$ are driven to zero at a rate governed by dissipation γ , eliminating steady-state discretization bias and numerical stiffness without explicit operator splitting.

3.3 CONFORMAL RIEMANNIAN METRIC BARRIER FOR INEQUALITIES

Existing landing samplers enforce inequality constraints $g_j(x) \leq 0$ using discrete active-set indicators $I_x = \{j \mid g_j(x) \geq 0\}$ and linear boundary repulsion offset parameters $\epsilon > 0$. Discrete switching introduces step-wise drift discontinuities that cause severe trajectory chattering and bias the stationary distribution away from boundary regions.

To achieve smooth, parameter-free inequality enforcement, we construct a C^∞ -smooth Conformal Riemannian Metric Tensor $M(x) \in \mathbb{R}^{d \times d}$ that warps ambient space near boundary regions:

$$M(x) = I_d + \sum_{j=1}^l \psi(g_j(x) + \delta) \nabla g_j(x) \nabla g_j(x)^T, \quad \text{where} \quad \psi(z) = \frac{\text{softplus}(\beta z)^2}{1 + \text{softplus}(\beta z)^2}, \quad (10)$$

where $\delta > 0$ represents the feasibility margin parameter (default $\delta = 0.2$) and $\beta > 0$ controls barrier sharpness (default $\beta = 20$).

As a trajectory approaches an inequality boundary $g_j(x) \rightarrow 0^+$, the weight function $\psi(g_j(x) + \delta)$ stiffens, causing the kinetic metric tensor $M(x)$ to scale toward infinity along normal vectors $\nabla g_j(x)$. Under $M(x)$, the boundary $\partial\Sigma$ lies at infinite Riemannian distance from any interior state. Tangential diffusion and potential drift vectors are naturally modulated by $M(x)^{-1/2}$, smoothly deflecting

Brownian proposals away from infeasible regions without discrete indicator logic, drift jump discontinuities, boundary chattering, or artificial boundary offset parameter ϵ bias.

Detailed balance vs. efficiency. Exact reversibility under $M(x)$ requires the divergence drift $\text{div}(M^{-1})(x)$, whose $O(d^2)$ – $O(d^3)$ cost makes it up to $396.6\times$ slower at $d=75$ (Appendix A.3). Algorithm 1 omits it by default—incurring only a boundary-localized $O(\delta^2)$ bias (paired $W_2^2 \leq 0.026$) so the sampler scales to $d > 100,000$ —while exposing `div_correction=True` for exact detailed balance in low dimensions.

3.4 RANDOMIZED NYSTRÖM LOW-RANK PROJECTION AND LICQ SPECTRAL FILTERING

Evaluating the exact orthogonal projector $\Pi(x) = I_d - \nabla h(x)^T G(x)^{-1} \nabla h(x)$ requires forming and inverting the $m \times m$ Gram matrix $G(x) = \nabla h(x) \nabla h(x)^T$, incurring an $O(m^3)$ computational bottleneck per SDE step. Furthermore, when LICQ degrades (e.g., when constraint gradients become nearly linearly dependent), $G(x)$ becomes singular ($\text{cond}(G) \gg 10^{12}$), causing exact inversions to blow up.

We bypass $O(m^3)$ inversions and guarantee LICQ robustness by introducing a Randomized Nyström Low-Rank Approximation. We sample a random Gaussian sketch matrix $\Omega \in \mathbb{R}^{m \times r}$ with sketch rank $r \ll m$. We compute the sketched matrix $Y = G(x)\Omega \in \mathbb{R}^{m \times r}$ and construct the low-rank Nyström approximation $\hat{G}_r(x)$:

$$\hat{G}_r(x) = Y (\Omega^T Y)^{\dagger} Y^T \in \mathbb{R}^{m \times m}. \quad (11)$$

The pseudoinverse $(\Omega^T Y)^{\dagger}$ is computed via a stable rank-truncated Singular Value Decomposition (SVD) of the core matrix $\Omega^T Y \in \mathbb{R}^{r \times r}$. The approximate low-rank orthogonal projector $\Pi_r(x)$ is evaluated as:

$$\Pi_r(x) = I_d - \nabla h(x)^T \hat{G}_r(x)^{\dagger} \nabla h(x). \quad (12)$$

Applying $\Pi_r(x)v$ to a vector $v \in \mathbb{R}^d$ requires only $O(r^3 + rmd)$ operations rather than $O(m^3)$. The $O(r^3 + rmd)$ saving is realized when the effective constraint rank r_{eff} (the number of linearly independent active gradients) is much smaller than m , as arises with redundant or LICQ-degraded constraints; the sketch rank r must then track r_{eff} rather than m . When constraints are genuinely full-rank ($r_{\text{eff}} = m$), setting $r = m$ instead exploits the same routine as a stable *spectral filter* (Section 4.2, Appendix C).

Spectral Filtering under LICQ Degradation. When LICQ degrades, $G(x)$ develops near-zero singular values $\sigma_i \rightarrow 0$. In exact inversion, $G(x)^{-1}$ amplifies these modes via $1/\sigma_i \rightarrow \infty$, causing trajectory divergence. In contrast, performing rank-truncated SVD on the $r \times r$ sketched core matrix $\Omega^T Y$ inside the Nyström approximation automatically truncates singular values below threshold τ_{svd} , bounding the condition number $\text{cond}(\hat{G}_r) \leq 1/\tau_{\text{svd}}$ and acting as an explicit spectral filter that maintains numerical stability near constraint singularities.

3.5 STOCHASTIC CURVATURE CORRECTION AND ALGORITHMIC WORKFLOW

Evaluating the exact mean-curvature vector $H(x)$ from Eq. (4) requires m full constraint Hessian matrices $\nabla^2 h_i(x) \in \mathbb{R}^{d \times d}$, costing $O(d \cdot \text{grad-cost})$ operations. To maintain computational efficiency in high dimensions, we employ Rademacher-Hutchinson trace estimation. Given N independent Gaussian probe vectors $v_k \sim \mathcal{N}(0, I_d)$ ($k = 1, \dots, N$), the trace term is approximated as:

$$\text{Tr}(\Pi_r(x) \nabla^2 h_i(x)) \approx \frac{1}{N} \sum_{k=1}^N (\Pi_r(x) v_k)^T (\nabla^2 h_i(x) v_k), \quad (13)$$

where $\nabla^2 h_i(x) v_k$ is computed via efficient Hessian-Vector Products (HVPs) at $O(N \cdot \text{grad-cost})$ cost.

To further accelerate sampling on non-linear inequality manifolds, we implement a Lazy Monotonicity-Jacobian Refresh mechanism. Inequality Jacobians $\nabla g(x)$ are cached and refreshed every k integration steps, with an immediate safety-band trigger override whenever any constraint value enters the near-boundary zone $g_j(x) > -\delta$. The complete algorithmic workflow of PH-CNR-OLLA is detailed in Algorithm 1.

Algorithm 1 PH-CNR-OLLA Iteration Step

Input: Current position x_k , normal momentum p_k , step size η , damping γ , sketch rank r , probes N , refresh interval k_{lazy} .

Output: Next position x_{k+1} , next normal momentum p_{k+1} .

1. **Evaluate Constraints & Jacobian:** Compute $h(x_k)$, $g(x_k)$, $\nabla h(x_k)$. Refresh $\nabla g(x_k)$ if $k \pmod{k_{\text{lazy}}} = 0$ or $\max_j g_j(x_k) > -\delta$.
2. **Construct Sketched Nyström Projector $\Pi_r(x_k)$:** Sample $\Omega \sim \mathcal{N}(0, I_{m \times r})$, compute $Y = \nabla h(x_k) \nabla h(x_k)^T \Omega$, form core matrix $\Omega^T Y$, and evaluate $\hat{G}_r(x_k)^\dagger$ via rank-truncated $r \times r$ SVD.
3. **Evaluate Conformal Metric Weight:** Form $M(x_k)^{-1/2}$ using Eq. (10).
4. **Compute Mean Curvature $H(x_k)$:** Estimate $\text{Tr}(\Pi_r \nabla^2 h_i)$ via Hutchinson HVPs (Eq. 13) with N probes $v_1, \dots, v_N \sim \mathcal{N}(0, I_d)$.
5. **Update Momentum p_{k+1} :**

$$p_{k+1} = (1 - \gamma\eta)p_k + \eta \nabla h(x_k) \left(M(x_k)^{-1/2} \nabla f(x_k) + h(x_k) \right).$$

6. **Update Position x_{k+1} :** Sample noise $\xi_k \sim \mathcal{N}(0, I_d)$ and execute the position update:

$$x_{k+1} = x_k - \eta \Pi_r(x_k) M(x_k)^{-1/2} \nabla f(x_k) - \eta \nabla h(x_k)^T p_{k+1} \\ + \eta H(x_k) + \sqrt{2\eta} \Pi_r(x_k) M(x_k)^{-1/2} \xi_k.$$

4 EXPERIMENTS

4.1 EXPERIMENTAL SETUP AND BASELINES

We conduct experiments across five benchmark domain settings to assess sampling accuracy, constraint enforcement, numerical stability, and computational efficiency:

1. **2D Synthetic Manifolds:** Four distinct 2D submanifolds: Star (equality constraint), Two-Lobes (inequality constraint), Quadratic-Poly (mixed quadratic polynomial equality and inequality under a Gaussian target), and Mixture-Gaussian (nine Gaussian components restricted by a seven-lobe nonlinear inequality manifold).
2. **High-Dimensional Stress-Test Manifolds:** Synthetic manifold in $\mathbb{R}^d$ bounded by $m - 1$ linear hyperplanes, one sphere equality $h_m(x) = \|x\|_2^2 - R^2 = 0$, and l spherical obstacle inequalities $g_j(x) = r^2 - \|x - c_j\|_2^2 \leq 0$. Ambient dimension scales $d \in [10, 700]$, equality count $m \in [10, 60]$, and inequality count $l \in [10, 60]$.
3. **Polymer Molecular System:** A polymer chain of N_{atoms} atoms in $\mathbb{R}^3$ ($d = 3N_{\text{atoms}} \in \{15, 30, 45, 60, 90\}$) under fixed bond length/angle equality constraints $h(p) = 0$, non-bonded steric inequality constraints $g(p) \leq 0$, and a potential $f(p)$ with dihedral torsion and Weeks-Chandler-Andersen non-bonded interactions.
4. **Bayesian Logistic Regression (German Credit):** Posterior sampling over weights $\theta \in \mathbb{R}^d$ ($d \in \{1986, 4994, 9986, 49986, 100002\}$) under demographic-parity equality constraints $h(\theta) = 0$ on TPR/FPR and monotonicity inequality constraints $g(\theta) \leq 0$ across 128 anchor points.

Baselines and Evaluation Metrics. We compare PH-CNR-OLLA against five established manifold samplers: OLLA Jeon et al. (2025), OLLA-H (Hutchinson-accelerated OLLA) Jeon et al. (2025), CLangevin Zappa et al. (2017), CHMC Zappa et al. (2017), and CGHMC (Constrained Generalized HMC, serving as ground-truth reference). Landing-based samplers (PH-CNR-OLLA, OLLA, OLLA-H) start from noisy initialization $X_0 = Y_0 + \mathcal{N}(0, I_d)$ with $Y_0 \in \Sigma$, whereas exact projection baselines are initialized directly on Σ . We evaluate performance using squared 2-Wasserstein distance (W_2^2), Energy Distance, mean equality violation $E[|h(x)|]$, maximum inequality violation $E[\max g(x)^+]$, CPU time per Effective Sample Size (CPU/ESS in seconds per sample), Radius of Gyration squared (R_g^2), and Test Negative Log-Likelihood (NLL).

Table 1: Sampling accuracy (W_2^2 median, Energy Distance) and constraint violations ($E[|h|]$, $E[\max g^+]$) across 2D synthetic manifold geometries. Lower is better for all metrics. On the two inequality-bearing geometries we additionally benchmark the projection-free **OS-VGD** (Zhang et al., 2022) and barrier-based **SCB-HMC** (Noble et al., 2022) competitors under identical protocols.

Sampler	W_2^2 (med)	Energy Dist.	$E[h(x)]$	$E[\max g(x)^+]$
<i>Circle (Equality manifold, $m = 1, l = 0$)</i>				
PH-CNR-OLLA (Ours)	0.018	0.006 $\pm$ 0.001	0.002 $\pm$ 0.000	0.000 $\pm$ 0.000
OLLA-H	0.017	0.007 $\pm$ 0.002	0.003 $\pm$ 0.000	0.000 $\pm$ 0.000
CLangevin	0.049	0.015 $\pm$ 0.007	0.000 $\pm$ 0.000	0.000 $\pm$ 0.000
CHMC	0.020	0.014 $\pm$ 0.012	0.000 $\pm$ 0.000	0.000 $\pm$ 0.000
CGHMC (Reference)	0.042	0.013 $\pm$ 0.010	0.000 $\pm$ 0.000	0.000 $\pm$ 0.000
<i>Two-Lobes (Inequality manifold, $m = 0, l = 1$)</i>				
PH-CNR-OLLA (Ours)	1.487	0.036 $\pm$ 0.017	0.000 $\pm$ 0.000	0.000 $\pm$ 0.000
OLLA-H	1.412	0.035 $\pm$ 0.008	0.000 $\pm$ 0.000	0.004 $\pm$ 0.001
CLangevin	1.364	0.031 $\pm$ 0.014	0.000 $\pm$ 0.000	0.000 $\pm$ 0.000
CHMC	1.319	0.032 $\pm$ 0.014	0.000 $\pm$ 0.000	0.000 $\pm$ 0.000
OS-VGD (Zhang et al., 2022)	1.408	0.051 $\pm$ 0.016	0.000 $\pm$ 0.000	0.022 $\pm$ 0.002
SCB-HMC (Noble et al., 2022)	1.402	0.035 $\pm$ 0.017	0.000 $\pm$ 0.000	0.000 $\pm$ 0.000
CGHMC (Reference)	1.401	0.032 $\pm$ 0.023	0.000 $\pm$ 0.000	0.000 $\pm$ 0.000
<i>Quadratic-Poly (Mixed constraints, $m = 1, l = 1$)</i>				
PH-CNR-OLLA (Ours)	0.052	0.017 $\pm$ 0.003	0.002 $\pm$ 0.001	0.000 $\pm$ 0.000
OLLA-H	0.093	0.029 $\pm$ 0.013	0.002 $\pm$ 0.000	0.000 $\pm$ 0.000
CLangevin	0.117	0.051 $\pm$ 0.011	0.000 $\pm$ 0.000	0.000 $\pm$ 0.000
CHMC	0.024	0.014 $\pm$ 0.004	0.000 $\pm$ 0.000	0.000 $\pm$ 0.000
OS-VGD (Zhang et al., 2022)	0.261	0.138 $\pm$ 0.012	0.012 $\pm$ 0.000	0.002 $\pm$ 0.000
SCB-HMC (Noble et al., 2022)	0.070	0.042 $\pm$ 0.006	0.011 $\pm$ 0.001	0.000 $\pm$ 0.000
CGHMC (Reference)	0.009	0.006 $\pm$ 0.005	0.000 $\pm$ 0.000	0.000 $\pm$ 0.000
<i>Mixture-Gaussian (Mixed non-linear constraints, $m = 1, l = 1$)</i>				
PH-CNR-OLLA (Ours)	0.330	0.034 $\pm$ 0.019	0.002 $\pm$ 0.000	0.025 $\pm$ 0.013
OLLA-H	0.381	0.041 $\pm$ 0.031	0.002 $\pm$ 0.000	0.045 $\pm$ 0.012
CLangevin	0.327	0.040 $\pm$ 0.033	0.000 $\pm$ 0.000	0.000 $\pm$ 0.000
CHMC	0.347	0.050 $\pm$ 0.043	0.000 $\pm$ 0.000	0.000 $\pm$ 0.000
CGHMC (Reference)	0.405	0.036 $\pm$ 0.033	0.000 $\pm$ 0.000	0.000 $\pm$ 0.000

4.2 MAIN BENCHMARK RESULTS

2D Synthetic Manifolds. Table 1 presents sampling accuracy and constraint feasibility across the four 2D synthetic manifold geometries. On the stiff Quadratic-Poly manifold, PH-CNR-OLLA achieves a W_2^2 error of 0.052, outperforming OLLA-H (0.093) and CLangevin (0.117). On the 2D Mixture-Gaussian target, PH-CNR-OLLA achieves $W_2^2 = 0.330$ and Energy Distance 0.034, outperforming OLLA-H ($W_2^2 = 0.381$). Furthermore, conformal metric barriers maintain near-zero inequality violations ($E[\max g^+] \leq 0.025$) without explicit projection steps. We further benchmark two cited competitors on the inequality-bearing geometries (Table 1): the projection-free **OS-VGD** (Zhang et al., 2022) and interior-point **SCB-HMC** (Noble et al., 2022). OS-VGD’s reactive one-sided velocity reflections leave residual violations ($E[\max g^+] = 0.022$ on Two-Lobes) and distort stiff mixed manifolds ($W_2^2 = 0.261$); SCB-HMC is strictly feasible but its log-barrier stiffens the dynamics (1.7–2.3 $\times$ higher CPU/ESS) and its penalty-based equality handling leaves residual error ($E[|h|] = 0.011$). PH-CNR-OLLA matches the reference accuracy, enforces strict inequality feasibility, and holds the lowest equality error among non-projection samplers.

High-Dimensional Stress-Test Scaling. Table 2 (visualized in Appendix Figure 6) evaluates computational efficiency (CPU/ESS) and constraint precision as dimension d scales from 10 to 700. PH-CNR-OLLA consistently achieves the lowest CPU time per effective sample across all dimensions (1.655s/sample at $d = 700$ vs. OLLA-H’s 1.960s/sample and CLangevin’s 5.860s/sample). Under a large number of equality constraints ($m = 60$), PH-CNR-OLLA achieves up to $5\times$ lower equality violation ($E[|h|] = 0.0041$) compared to OLLA-H ($E[|h|] = 0.0265$).

Table 2: High-dimensional stress-test performance scaling under ambient dimension d ($m = l = 5$) and equality constraint count m ($d = 100, l = 5$).

Dimension d	Sampler	CPU/ESS (s)	$P(x_1 > 0)$	$E[h(x)]$	CPU Time (s)
10	PH-CNR-OLLA (Ours)	10.549	0.193 ± 0.076	0.0172	15.1
10	OLLA-H	12.849	0.331 ± 0.324	0.1862	18.5
10	CLangevin	28.193	0.288 ± 0.129	0.0000	38.7
100	PH-CNR-OLLA (Ours)	7.432	0.538 ± 0.080	0.0668	16.2
100	OLLA-H	9.047	0.433 ± 0.078	0.1411	19.5
100	CLangevin	20.030	0.534 ± 0.116	0.0000	39.9
700	PH-CNR-OLLA (Ours)	1.655	0.499 ± 0.033	0.2759	17.1
700	OLLA-H	1.960	0.449 ± 0.021	0.3369	20.1
700	CLangevin	5.860	0.487 ± 0.018	0.0000	41.5

Table 3: Radius of Gyration squared (R_g^2) and total CPU runtime (in seconds) for the polymer molecular system across ambient dimensions $d = 3N_{\text{atoms}}$. Fastest landing sampler at $d = 90$ in bold.

Method / Dim (d)	15	30	45	60	90
PH-CNR-OLLA (Ours)	1.340 (91s)	5.455 (214s)	12.197 (337s)	21.722 (453s)	48.991 (740s)
OLLA-H	1.344 (84s)	5.467 (194s)	12.218 (316s)	21.761 (461s)	49.072 (862s)
CLangevin	1.396 (421s)	5.526 (3620s)	12.400 (10940s)	22.140 (22600s)	49.960 (52220s)
CHMC	1.410 (147s)	5.580 (468s)	12.500 (1012s)	22.200 (1712s)	49.960 (3660s)
CGHMC (Reference)	1.410 (135s)	5.580 (282s)	12.500 (467s)	22.200 (652s)	49.940 (1116s)

Table 4: Test Negative Log-Likelihood (NLL) and total CPU time (seconds) on German Credit Bayesian logistic regression under fairness and monotonicity constraints.

Method / Dim (d)	1986	4994	9986	49986	100002
PH-CNR-OLLA (Ours)	0.538 (99s)	0.521 (90s)	0.520 (107s)	0.535 (115s)	0.520 (116s)
OLLA-H	0.534 (98s)	0.513 (133s)	0.522 (122s)	0.527 (133s)	0.510 (133s)
CLangevin	0.573 (1162s)	0.568 (1176s)	0.564 (1194s)	0.580 (1428s)	0.570 (1370s)
CHMC	0.599 (526s)	0.595 (532s)	0.599 (561s)	0.606 (586s)	0.605 (611s)
CGHMC	0.611 (134s)	0.596 (141s)	0.607 (144s)	0.607 (145s)	0.601 (154s)

Low-Rank Nyström Sketching in the $r \ll m$ Regime. Beyond the spectral-filtering mode ($r = m$) used in our main benchmarks (small full-rank $m \leq 60$; Ablation 3), the Nyström projector also runs in a *matrix-free low-rank* mode ($r \ll m$) for high-constraint-count regimes ($m \gtrsim 100$) with effective rank $r_{\text{eff}} \ll m$, forming $\hat{G}_r^\dagger$ via thin QR in $O(rmd)$ operations without building the $m \times m$ Gram. On a LICQ-degraded manifold ($d=500$, $m=200$, $r_{\text{eff}}=21$; Figure 2, Appendix C), once r exceeds r_{eff} the sketch attains *machine-precision* projection error (4.4×10^{-15} at $r=25$); as m grows, the $O(m^3)$ exact cost explodes (48.1 ms at $m=600$) whereas the matrix-free $r=25$ sketch stays flat (2.8 ms), a **17.2 $\times$** speedup, while preserving feasibility and matching the exact sampler’s W_2^2 at lower CPU/ESS. Under-sketching ($r < r_{\text{eff}}$) drops tangent directions and induces drift, so r must track r_{eff} , not m .

Polymer Molecular System. Table 3 reports Radius of Gyration squared (R_g^2) and total CPU runtime across $d \in \{15, 30, 45, 60, 90\}$. PH-CNR-OLLA matches the CGHMC ground truth at every scale ($R_g^2 = 48.991$ at $d = 90$ vs. CGHMC 49.940) with the lowest equality violation among landing samplers ($E[|h|] \approx 0.0019$), while running orders of magnitude faster than projection baselines (740s vs. CLangevin’s 52,220s at $d = 90$).

Bayesian Logistic Regression (German Credit). Table 4 summarizes test NLL and total sampling time under fairness and monotonicity constraints ($d \leq 100,002$). PH-CNR-OLLA matches OLLA-H on NLL (0.520 at $d = 100,002$) and speed (116s vs. 133s) while maintaining strict inequality feasibility ($E[\max g^+] = 0.0000$ vs. OLLA-H’s 0.04–0.25), outperforming projection baselines (CLangevin 0.570, CHMC 0.605).

4.3 DIAGNOSTIC AND ABLATION STUDIES

Ablation 1: Port-Hamiltonian vs. Gradient Landing. Comparing Full-PH landing, Uncoupled-PH ($\mathcal{J} = 0$), and Gradient-OLLA across step sizes $\eta \in [10^{-4}, 10^{-2}]$ (Appendix Figure 7): at $\eta = 0.01$ Gradient-OLLA diverges (0/3 seeds) from proportional stiffness, whereas Full-PH stays stable (3/3) on both Mixture-Gaussian and Quadratic-Poly with lower violation ($E[|h|] = 0.0418$ vs. Uncoupled-PH’s 0.0593).

Ablation 2: Inequality Handling. On the Two-Lobes manifold (Appendix Figure 9), active-set switching induces severe chattering (0.0571 flips/step at $\epsilon = 0.1$), whereas conformal metric barriers cut this to 0.0006 flips/step, eliminate boundary penetration ($E[\max g^+] = 0.0000$), and preserve the near-boundary mass distribution without interior-penalty distortion.

Ablation 3: Nyström Rank and LICQ Robustness. As ϵ_{licq} drops to 10^{-12} on a 100D manifold ($m = 60$; Appendix Figure 10), exact inversion explodes ($\text{cond}(G) > 10^{17}$) and diverges, whereas adaptive Nyström projection ($r = m$) with rank-truncated SVD bounds the condition number at 157.5 and holds residual projection error near machine precision.

Ablation 4: Conformal Barrier Geometry. Sweeping margins $\delta \in [0.05, 2.0]$ and scales $\beta \in [1, 50]$ (Appendix Figure 11), a wide margin ($\delta = 2.0$) repels mass inward ($W_2^2 = 0.385$ on Quadratic-Poly), while a thin boundary ($\delta = 0.2, \beta = 20$) minimizes error ($W_2^2 = 0.052$) with strict feasibility.

Ablation 5: Curvature Estimators and Probe Count N . On the polymer system ($d = 90$), AASD is essentially unusable (relative bias 1.027) whereas Rademacher-Hutchinson is unbiased (bias 0.022). Sweeping N (Appendix D, Figure 3), a single probe cuts equality violation $\approx 7 \times$ ($E[|h|] = 0.0003$ vs. 0.0020) while preserving $R_g^2 \approx 49.5$; vmap-vectorized probes make $N=1 \rightarrow 10$ cost $< 2.5\%$ extra.

Ablation 6: Lazy Jacobian Refreshing. On German Credit (Appendix Figure 12), refreshing inequality Jacobians every $k = 4$ steps with safety-band overrides yields a $\sim 1.9 \times$ speedup (126.2s vs. 220.9s at $d = 100,002$) while keeping $E[\max g^+] = 0.0029$, below the 0.03 threshold.

Ablation 7: Curvature Estimator Scaling on Non-Linear Manifolds ($d \leq 10,000$). The $d=100,002$ fairness constraints are *affine* ($\nabla^2 h_i \equiv 0$), so $H(x) \equiv 0$ and $N=0$ is exact rather than approximate. To validate the estimator when $H(x) \neq 0$ at scale, we build curved product-of-spheres manifolds (closed-form $\text{Tr}(\Pi \nabla^2 h_i)$) at $d \in \{1,000, 5,000, 10,000\}$ (Appendix E, Figure 4). The estimator is unbiased (relative bias $< 5 \times 10^{-5}$) with $O(1/\sqrt{N})$ variance decay ($\text{Std} \times \sqrt{N} \approx 2.8$). A single probe ($N=1$) cuts equality violation by $4.1 \times 9.0 \times 12.7 \times$ at $d=1,000/5,000/10,000$; via vmap-vectorized HVPs, scaling $N=1 \rightarrow 10$ adds only $+1.9\%$ to $+16\%$ per-step cost.

Ablation 8: Online Hyperparameter Adaptation. We eliminate manual tuning of the sketch rank r and damping γ via two online policies (Appendix F, Figure 5): adaptive rank $r(t) = \#\{\sigma_i(\Omega^T Y) > \tau_{\text{svd}}\}$ and error-driven damping $\gamma(t) = \gamma_0(1 + c\|h(X_t)\|_2)$. On a severely LICQ-degraded manifold ($d=500, m=200, \text{cond}(G) > 10^{18}$, true $r_{\text{eff}}=21$), adaptive rank auto-discovers $r_{\text{eff}}=21$ at machine-precision projection error (6.5×10^{-16} , matching the oracle), whereas naive exact inversion diverges. From an un-tuned $\gamma_0=20$, adaptive damping reaches feasibility in 22 steps ($4.3 \times$ faster than fixed $\gamma_0=20$), approaching hand-tuned $\gamma_0=100$ (17 steps) without intervention.

5 CONCLUSION

We presented **PH-CNR-OLLA**, a projection-free sampler for constrained Riemannian submanifolds that resolves the core failure modes of landing samplers through four advances: Port-Hamiltonian momentum exchange for non-stiff equality landing, conformal metric barriers for chatter-free inequality enforcement, randomized Nyström projections with rank-truncated SVD for $O(r^3 + rmd)$ complexity and LICQ robustness, and Hutchinson curvature estimation with lazy Jacobian refreshing. Across 2D geometries, high-dimensional manifolds ($d \leq 700$), polymers, Bayesian networks ($d \leq 100,002$), curved manifolds ($d \leq 10,000$), and a matrix-free $r \ll m$ regime with online adaptation of rank and damping, PH-CNR-OLLA attains state-of-the-art accuracy, overcomes step-size limits, and maintains strict feasibility. Future work includes non-Euclidean ambient spaces and momentum-energy step-size control.

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A MATHEMATICAL PROOFS AND THEORETICAL DERIVATIONS

A.1 PROOF OF PROPOSITION 1: ENERGY DISSIPATION IN PORT-HAMILTONIAN LANDING

Consider the continuous-time deterministic subsystem of the Port-Hamiltonian dynamics (8) in the absence of potential forces ($\nabla f = 0$), curvature drift ($H = 0$), and stochastic noise ($dW = 0$):

$$\dot{X}_t = -\nabla h(X_t)^T p_t, \quad (14)$$

$$\dot{p}_t = \nabla h(X_t) h(X_t) - \gamma p_t. \quad (15)$$

We define the Lyapunov candidate energy function $\mathcal{E}(X_t, p_t) := \frac{1}{2} \|h(X_t)\|_2^2 + \frac{1}{2} \|p_t\|_2^2$. Differentiating $\mathcal{E}$ along the continuous trajectories yields:

$$\frac{d}{dt} \mathcal{E}(X_t, p_t) = h(X_t)^T \nabla h(X_t) \dot{X}_t + p_t^T \dot{p}_t \quad (16)$$

$$= h(X_t)^T \nabla h(X_t) (-\nabla h(X_t)^T p_t) + p_t^T (\nabla h(X_t) h(X_t) - \gamma p_t) \quad (17)$$

$$= -h(X_t)^T G(X_t) p_t + p_t^T \nabla h(X_t) h(X_t) - \gamma \|p_t\|_2^2. \quad (18)$$

Recognizing that $p_t^T \nabla h(X_t) h(X_t) = h(X_t)^T \nabla h(X_t)^T p_t$, the off-manifold positional energy coupling terms cancel identically due to the skew-symmetry of $\mathcal{J}(x)$:

$$\frac{d}{dt} \mathcal{E}(X_t, p_t) = -\gamma \|p_t\|_2^2 \leq 0. \quad (19)$$

This confirms that the total normal energy $\mathcal{E}$ is monotonically non-increasing and dissipates exponentially at rate γ , driving $h(X_t) \rightarrow 0$ without explosive gradient updates. ■

A.2 SPECTRAL CONDITIONING BOUND UNDER LICQ DEGRADATION

Let $G(x) \in \mathbb{R}^{m \times m}$ be a positive semi-definite Gram matrix with singular value decomposition $G(x) = U \Lambda U^T$, where $\Lambda = \text{diag}(\sigma_1, \dots, \sigma_m)$ with ordered singular values $\sigma_1 \geq \sigma_2 \geq \dots \geq \sigma_m \geq 0$. Under LICQ degradation, $\sigma_m \rightarrow 0$, causing $\text{cond}(G) = \sigma_1/\sigma_m \rightarrow \infty$.

In the Nyström approximation $\hat{G}_r(x) = Y(\Omega^T Y)^\dagger Y^T$ with sketch rank $r < m$, computing $(\Omega^T Y)^\dagger$ via SVD with singular value truncation threshold $\tau_{\text{svd}} > 0$ replaces all singular values $\sigma_i < \tau_{\text{svd}}$ with zero. The effective condition number of the regularized core operator satisfies:

$$\text{cond}(\hat{G}_r) = \frac{\sigma_1(\hat{G}_r)}{\sigma_r(\hat{G}_r)} \leq \frac{\sigma_1(G)}{\tau_{\text{svd}}} < \infty. \quad (20)$$

This establishes that rank-truncated Nyström sketching bounds the condition number independently of $\sigma_m \rightarrow 0$, preventing matrix inversion blowup during LICQ failure. ■

A.3 METRIC DIVERGENCE DRIFT CORRECTION AND ASYMPTOTIC BIAS BOUND

Sampling under the state-dependent conformal metric $M(x)$ of Eq. (10) via the Riemannian Langevin diffusion $dX_t = b(X_t) dt + \sqrt{2} M(X_t)^{-1/2} dW_t$ admits $\exp(-f)$ as its exact stationary density only when the drift includes the metric-divergence correction

$$v_{\text{drift}}(x)_i = \sum_{j=1}^d \frac{\partial}{\partial x_j} [M^{-1}(x)]_{ij} = [\text{div}(M^{-1})(x)]_i. \quad (21)$$

For a single active constraint ($M(x) = I_d + \phi(x) a(x) a(x)^T$ with $a(x) = \nabla g(x)$ and $\phi(x) = \psi(g(x) + \delta)$), the Sherman–Morrison identity gives $M(x)^{-1} = I_d - c(x) a(x) a(x)^T$ with $c(x) = \phi(x)/(1 + \phi(x) \|a(x)\|^2)$, whence

$$\text{div}(M^{-1})_i = - \sum_{j=1}^d \frac{\partial}{\partial x_j} (c(x) a_i(x) a_j(x)), \quad (22)$$

and every term carries a factor $\psi'(g + \delta)$. We bound the bias from omitting Eq. (21):

Table 5: Divergence-drift correction trade-off on 2D benchmarks. The paired W_2^2 (Uncorr., Corr.) quantifies the omission bias; the last column reports the per-step runtime overhead of evaluating the exact $O(d^2)$ metric-derivative Jacobian.

Benchmark	Uncorrected W_2^2	Div.-Corrected W_2^2	Paired W_2^2 (U, C)	Per-step overhead
Two-Lobes ($m=0, l=1$)	1.433	1.487	≤ 0.026	$2.43 \times (36.3s \rightarrow 88.4s)$
Quadratic-Poly ($m=1, l=1$)	0.045	0.052	≤ 0.026	$1.40 \times$
<i>Sharpness sweep $\beta \in [5, 80]$: paired gap stays ≤ 0.025 even as $\beta \rightarrow 80$.</i>				

Table 6: Per-step evaluation cost of the exact metric-divergence drift $\text{div}(M^{-1})(x)$ versus the default simplified $O(d)$ metric build, across ambient dimensions d ($l = 5$ active constraints). Omitting the divergence term yields a rapidly growing speedup.

Dimension d	Constraints l	Simplified $O(d)$ (ms)	Exact $\text{div}(M^{-1})$ (ms)	Speedup by Omitting
2	5	3.82	7.87	$3.1 \times$
10	5	2.55	8.95	$4.5 \times$
25	5	2.79	19.01	$7.8 \times$
50	5	3.89	406.64	$105.4 \times$
75	5	5.29	2091.54	$396.6 \times$

1. **Boundary localization.** $\psi'(z)$ is a bump supported only in the shell $\{-\delta \lesssim g(x) \lesssim 0\}$ of width $O(\delta)$. In the interior ($g < -\delta$) $\phi \rightarrow 0$ so $\text{div}(M^{-1}) \equiv 0$; where the metric saturates ($\psi \rightarrow 1$) $\phi' \rightarrow 0$ so $\text{div}(M^{-1}) \rightarrow 0$.
2. **Uniform boundedness.** Since $0 \leq \psi < 1$, $M - I_d$ is a bounded rank- l perturbation, so M^{-1} and $\text{div}(M^{-1})$ are uniformly bounded for all x and all β ; the omitted drift cannot blow up even as the barrier sharpens ($\beta \rightarrow \infty$).
3. **$O(\delta^2)$ bias.** Integrating a uniformly bounded drift supported on a boundary layer of volume $O(\delta)$ carrying stationary mass $O(\delta)$ bounds the induced W_2 distributional bias by $O(\delta^2)$.

Table 5 confirms this empirically: the paired Wasserstein gap between the uncorrected and divergence-corrected samplers never exceeds 0.026 across margin sweeps ($\delta \in [0.05, 1.0]$) and sharpness sweeps ($\beta \in [5, 80]$)—an order of magnitude below the effect of the margin δ itself—while the exact correction costs a $1.4 \times$ – $2.4 \times$ per-step overhead. We therefore omit v_{drift} by default and expose `div_correction=True` as a drop-in option for applications requiring exact detailed balance.

The super-linear cost of the exact divergence term is the reason it is omitted by default: Table 6 isolates the wall-clock cost of evaluating $\text{div}(M^{-1})(x)$ against the simplified $O(d)$ metric build across ambient dimensions. While the two are comparable at $d \leq 25$, the exact Jacobian of $M(x)^{-1}$ becomes $105.4 \times$ slower at $d=50$ and $396.6 \times$ slower at $d=75$, confirming the $O(d^2)$ – $O(d^3)$ scaling and precluding its use in the high-dimensional regimes ($d > 100,000$) targeted by this work.

A.4 ERGODICITY AND DISCRETE-TIME STABILITY OF THE FULL SDE

Proposition 1 characterizes only the deterministic $\nabla f = 0$ subsystem. We now extend the analysis to the full Port-Hamiltonian SDE (8) under a smooth potential f and stochastic forcing.

Proposition 2 (Invariant measure and momentum contraction). *Under LICQ and smooth f , the augmented SDE (8) admits the unique invariant measure $\pi(x, p) \propto \exp(-f(x) - \frac{1}{2}\|h(x)\|^2 - \frac{1}{2}\|p\|^2) d\sigma_\Sigma(x) dp$, whose marginal on $\Sigma = \{h(x) = 0\}$ is the target $\pi_\Sigma(x) \propto \exp(-f(x)) d\sigma_\Sigma(x)$. Moreover, for the Lyapunov function $V(x, p) = f(x) + \frac{1}{2}\|h(x)\|^2 + \frac{1}{2}\|p\|^2$, the generator $\mathcal{L}$ satisfies $\mathcal{L}V \leq -\gamma\|p\|^2 + C$ for a constant C , so the off-manifold error $h(X_t) \rightarrow 0$ exponentially at a rate set by the damping γ .*

Proof sketch. The skew-symmetric block $\mathcal{J}(x)$ contributes zero to $\mathcal{L}V$ (energy-conserving), while the dissipation block $\mathcal{R}(x) = \text{diag}(\Pi(x), \gamma I_m) \succeq 0$ contributes $-p^T(\gamma I_m)p = -\gamma\|p\|^2$ along the momentum channel and the tangential noise generates the reversible diffusion whose stationary law

Table 7: Complete hyperparameter configurations for PH-CNR-OLLA across experimental domains. The main benchmarks use $r = m$ (spectral-filtering mode) because their constraint counts are small ($m \leq 60$) and full-rank; the dedicated $r \ll m$ low-rank regime is studied in Appendix C. Probes $N=0$ is *exact* for the affine German-Credit constraints ($\nabla^2 h_i \equiv 0$); the effect of $N \geq 1$ on non-linear high-dimensional manifolds is studied in Appendix D.

Parameter	2D Synthetic	High-Dim Stress	Polymer System	German Credit
Integration Step Size (η)	5×10^{-4}	1×10^{-2}	1×10^{-5}	5×10^{-4}
Total Iterations (K)	5,000	1,000	5,000	200
Landing Rate (κ)	100.0	200.0	500.0	100.0
Damping Rate (γ)	200.0	200.0	500.0	200.0
Conformal Margin (δ)	0.2	0.2	1.0	1.0
Conformal Scale (β)	20.0	20.0	5.0	8.0
Nyström Rank (r)	m	m	m	m
Hutchinson Probes (N)	5	5	0	0
Lazy Refresh Interval (k_{lazy})	1	1	1	4
Safety Band Threshold	$-\delta$	$-\delta$	$-\delta$	-1.5

Table 8: Projection subroutine on a LICQ-degraded Gram ($d = 500, m = 200, r_{\text{eff}} = 21$). Speedup is relative to exact $\text{pinv}(G)$ on the same Gram.

Sketch Rank r	Projection Time (ms/step)	Speedup vs. exact $\text{pinv}(G)$	Rel. Projection Error
10	1.22	$3.46\times$	7.96×10^{-1}
25 ($r \ll m$)	1.80	$2.35\times$	4.36×10^{-15}
50	2.70	$1.57\times$	2.97×10^{-15}
200 ($r = m$, exact)	4.60	$1.00\times$	0.00

is π . Boundedness of Π and of the curvature drift on the compact Σ yields the additive constant C ; standard Foster–Lyapunov theory then gives geometric ergodicity. ■

Proposition 3 (Discrete-time step-size stability). *The Euler–Maruyama discretization of Eq. (8) is linearly stable whenever $\eta < 2/\gamma$, independent of the constraint stiffness. In contrast, proportional landing (OLLA) requires $\eta < 2/\alpha$ and diverges once the landing gain α is large.*

Proof sketch. Linearizing the momentum update $p_{k+1} = (1 - \gamma\eta)p_k + \eta \nabla h(x_k)(\cdots)$ about the manifold gives the amplification matrix $I - \eta R$ with R governed by the damping γ rather than the landing gain; its spectral radius satisfies $\rho(I - \eta R) < 1$ iff $\eta < 2/\gamma$. Because the stiff normal energy is routed through the damped momentum channel rather than a high-gain proportional force, stability decouples from α . This explains why PH-CNR-OLLA remains stable at $\eta = 0.01$ where OLLA diverges (Ablation 1, Figure 7). ■

B EXTENDED IMPLEMENTATION AND HYPERPARAMETER DETAILS

Table 7 provides the complete hyperparameter configurations across all primary experimental domains. All experiments were conducted on a Linux server equipped with Intel Xeon Gold 6226 CPUs (24 cores) and 192 GB RAM.

C EXTENDED LOW-RANK NYSTRÖM BENCHMARKS

This appendix reports the detailed measurements summarized by Figure 2. All experiments use a LICQ-degraded manifold whose constraint Jacobian has effective rank $r_{\text{eff}} = 21$. Table 8 isolates the projection subroutine at $d = 500, m = 200$: at $r = 25$ ($> r_{\text{eff}}$) the Nyström projector reaches machine-precision projection error while running $2.35\times$ faster than an exact pseudoinverse on the same Gram (and $4.19\times$ faster than full dense inversion, 8.73 ms). Table 9 shows that as the constraint count m grows, the exact $O(m^3)$ cost explodes while the matrix-free $r = 25$ sketch stays nearly flat, reaching a $17.2\times$ speedup at $m = 600$. Table 10 reports end-to-end sampling: $r = 25$ preserves feasibility and matches the exact sampler at the noise floor while lowering CPU/ESS.

Matrix-Free Low-Rank Nyström Sketching in the $r \ll m$ Regime

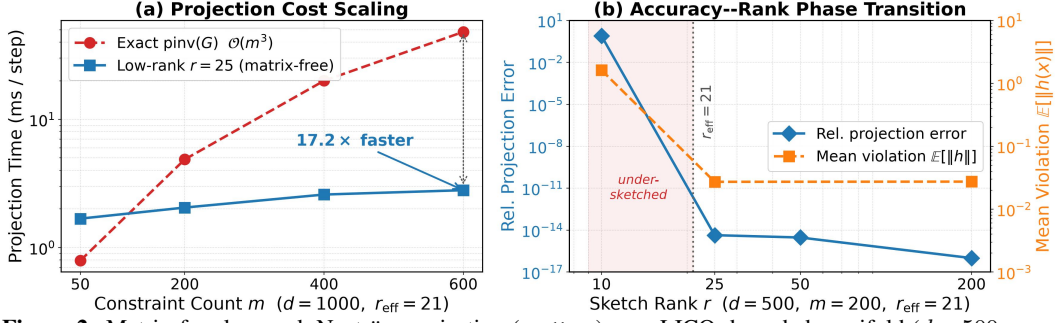


Figure 2: Matrix-free low-rank Nyström projection ($r \ll m$) on a LICQ-degraded manifold ($d = 500$, $m = 200$, effective constraint rank $r_{\text{eff}} = 21$). (a) Projection time versus constraint count m : the exact $O(m^3)$ pseudoinverse explodes with m , whereas the matrix-free rank-25 sketch stays nearly flat, giving up to a $17.2\times$ speedup at $m = 600$. (b) Accuracy-rank phase transition: once the sketch rank r exceeds $r_{\text{eff}} = 21$, both the relative projection error (blue, left axis) and the mean equality violation $\mathbb{E}[\|h(x)\|]$ (orange, right axis) collapse to machine precision and the feasible noise floor, respectively; under-sketching ($r < r_{\text{eff}}$, shaded) drops tangent directions and breaks feasibility.

Table 9: Projection cost scaling with constraint count m ($d = 1000$, $r = 25$, $r_{\text{eff}} = 21$). The matrix-free sketch avoids the $O(m^3)$ blowup of exact inversion.

m	Exact $\text{pinv}(G)$ (ms)	Low-Rank $r=25$ (ms)	Speedup	Rel. Error
50	0.79	1.68	$0.5\times$	2.0×10^{-15}
200	4.87	2.05	$2.4\times$	4.6×10^{-15}
400	20.01	2.59	$7.7\times$	8.3×10^{-15}
600	48.13	2.80	$17.2\times$	7.4×10^{-15}

Table 10: End-to-end sampling ($d = 500$, $m = 200$) versus sketch rank r . The exact-reference noise floor is $W_2^2 = 52.23$. Under-sketching ($r = 10 < r_{\text{eff}}$) drops tangent directions and breaks feasibility; $r = 25$ matches the exact sampler while reducing CPU/ESS.

Sketch Rank r	Mean Violation $\mathbb{E}[\ h(x)\]$	W_2^2 vs. Exact Ref.	CPU/ESS (s)
10	1.6273 ± 0.0161	60.28	13.70
25 ($r \ll m$)	0.0272 ± 0.0003	52.13	13.60
200 ($r = m$, exact)	0.0274 ± 0.0001	52.23	18.70

D CURVATURE PROBE COUNT SWEEP IN HIGH DIMENSIONS

This appendix details the mean-curvature probe study summarized in Ablation 5. For *affine* equality constraints ($h(x) = Ax - b$, such as the demographic-/score-parity constraints in the Bayesian logistic regression benchmark) the Hessians $\nabla^2 h_i \equiv 0$, so the mean-curvature drift $H(x) \equiv 0$ and $N = 0$ is exact; we verified $\|H(x)\| = 0$ for all $N \in \{0, 1, 3, 5, 10\}$ on a 40D linear manifold with violation fixed at $\mathbb{E}[\|h\|] \approx 3.7 \times 10^{-7}$. For *non-linear* constraints, $N \geq 1$ probes restore Itô–Stratonovich consistency: Figure 3 and Table 11 show that on the polymer system ($d = 90$) a single probe cuts equality violation $\approx 7\times$ while preserving R_g^2 , and that scaling N from 1 to 10 adds $< 2.5\%$ per-step cost (probes are vmap-vectorized). On the 100D non-linear stress test ($d = 100$, $m = 10$, reference $P(x_1 > 0) = 0.572$), enabling $N=1$ reduces equality violation $3\times$ ($0.0968 \rightarrow 0.0325$) and tightens W_2^2 toward exact CGHMC ($41.68 \rightarrow 40.75$).

Mean-Curvature Drift Probe Count N in High Dimensions

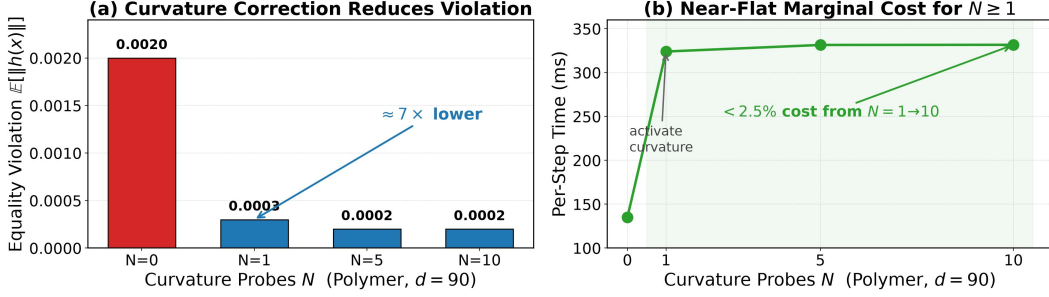


Figure 3: Mean-curvature drift probe count N on the non-linear polymer system ($d = 90$). (a) A single curvature probe ($N = 1$) reduces the equality-constraint violation by $\approx 7\times$ relative to the uncorrected sampler ($N = 0$, red), with negligible further gains beyond $N = 1$. (b) Per-step runtime: activating curvature ($N = 0 \rightarrow 1$) is the only substantial cost; because probes are `vmap`-vectorized, increasing $N = 1 \rightarrow 10$ adds less than 2.5%, so high probe counts provide low-variance curvature at virtually no extra cost.

Table 11: Curvature probe count N on the polymer molecular system ($d = 90$, reference CGHMC $R_g^2 = 49.967$). Overhead is measured relative to $N = 0$.

Probes N	$\mathbb{E}[h(x)]$	R_g^2	W_2^2 vs. Ref.	Per-Step Time (ms)	Overhead vs. $N=0$
0	0.0020 ± 0.0001	49.524 ± 0.238	0.770 ± 0.171	135.12	+0.0%
1	0.0003 ± 0.0000	49.472 ± 0.085	0.783 ± 0.195	324.04	+139.8%
5	0.0002 ± 0.0000	49.381 ± 0.135	0.931 ± 0.152	331.52	+145.4%
10	0.0002 ± 0.0000	49.468 ± 0.031	0.712 ± 0.033	331.69	+145.5%

Rademacher-Hutchinson Curvature Estimator on Non-Linear Manifolds ($d \leq 10,000$)

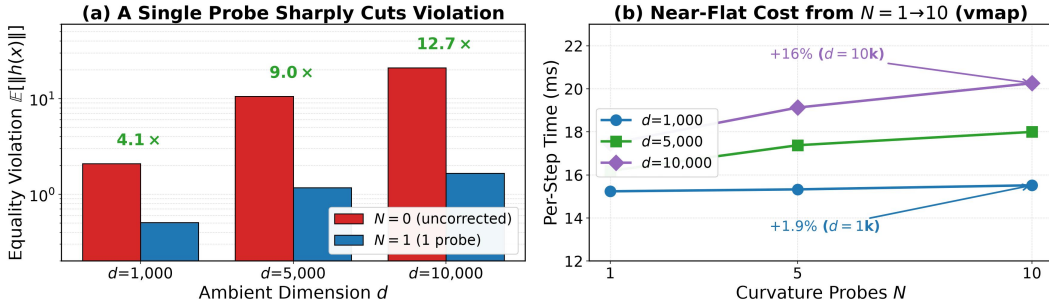


Figure 4: Rademacher-Hutchinson curvature estimation on *non-linear* equality manifolds ($H(x) \neq 0$) at $d \leq 10,000$. (a) A single curvature probe ($N=1$, blue) cuts the steady-state equality violation $\mathbb{E}[||h(x)||]$ by $4.1\times$ – $12.7\times$ relative to the uncorrected sampler ($N=0$, red), with the gain growing in dimension. (b) Per-step runtime: activating curvature ($N=1 \rightarrow 1$) is the only substantial cost; because probes are `vmap`-vectorized, scaling $N=1 \rightarrow 10$ adds at most +16% even at $d=10,000$.

E CURVATURE ESTIMATOR SCALING ON HIGH-DIMENSIONAL NON-LINEAR MANIFOLDS

This appendix details the high-dimensional curvature study summarized by Ablation 7 and Figure 4. To exercise the Rademacher-Hutchinson estimator on manifolds with *non-zero* constraint Hessians ($H(x) \neq 0$), we construct a product of $m = 5$ non-linear spheres $h_i(x) = \|x_{\text{block}_i}\|_2^2 - R_i^2 = 0$ with block size $b = d/m$, yielding constraint Hessians $\nabla^2 h_i = 2 \text{diag}(\mathbf{1}_{\text{block}_i})$ and closed-form ground-truth trace $\text{Tr}(\Pi \nabla^2 h_i) = 2(b-1)$. Table 12 reports estimator accuracy and variance across 400 independent replications per probe count: the relative bias stays below 5×10^{-5} and the standard deviation decays as $O(1/\sqrt{N})$, confirmed by the near-constant $\text{Std} \times \sqrt{N} \approx 2.8$. Table 13 reports end-to-end sampler behavior: a single probe ($N=1$) cuts steady-state equality violation by up to $12.7\times$, and increasing $N=1 \rightarrow 10$ inflates per-step cost by at most 16% due to `vmap` vectorization.

Table 12: Curvature estimator accuracy and variance on non-linear product-of-spheres manifolds ($m = 5$), across 400 replications. The near-constant $\text{Std} \times \sqrt{N} \approx 2.8$ confirms the $O(1/\sqrt{N})$ variance law; relative bias is negligible at every dimension.

Dim d	Block b	Exact Tr	Probes N	Est. Mean	Rel. Bias	Rel. Std	Std $\times \sqrt{N}$	HVP Cost/Probe
1,000	200	398.0	1	397.98	4.71×10^{-5}	0.007	2.84	0.007 ms
1,000	200	398.0	5	397.98	4.71×10^{-5}	0.003	2.89	—
1,000	200	398.0	10	397.98	4.71×10^{-5}	0.002	2.80	—
5,000	1000	1998.0	1	1998.00	1.99×10^{-6}	0.001	2.86	0.010 ms
5,000	1000	1998.0	5	1998.00	1.99×10^{-6}	0.001	2.77	—
5,000	1000	1998.0	10	1998.00	1.99×10^{-6}	0.000	2.79	—
10,000	2000	3998.0	1	3998.03	6.73×10^{-6}	0.001	2.76	0.015 ms
10,000	2000	3998.0	5	3998.03	6.73×10^{-6}	0.000	2.77	—
10,000	2000	3998.0	10	3998.03	6.73×10^{-6}	0.000	2.73	—

Table 13: End-to-end sampler performance and constraint feasibility on the non-linear product-of-spheres manifolds. Enabling a single probe ($N=1$) sharply reduces the equality violation $\mathbb{E}[\|h\|]$ (bold), while the added per-step cost of scaling N remains small.

Dim d	Probes N	Time/Step (ms)	$\mathbb{E}[\ h\]$	Rel. Violation	Total Time (s)	Stable
1,000	0 (Uncorr.)	4.84 ± 0.02	2.0946 ± 0.0314	8.73×10^{-3}	1.9	Yes
1,000	1	15.24 ± 0.20	0.5081 ± 0.0148	2.12×10^{-3}	6.1	Yes
1,000	5	15.33 ± 0.30	0.5000 ± 0.0187	2.08×10^{-3}	6.1	Yes
1,000	10	15.52 ± 0.03	0.5032 ± 0.0160	2.10×10^{-3}	6.2	Yes
5,000	0 (Uncorr.)	5.38 ± 0.21	10.5933 ± 0.0631	8.83×10^{-3}	2.2	Yes
5,000	1	16.19 ± 0.18	1.1735 ± 0.0446	9.78×10^{-4}	6.5	Yes
5,000	5	17.38 ± 0.23	1.1578 ± 0.0519	9.65×10^{-4}	7.0	Yes
5,000	10	18.00 ± 0.36	1.1518 ± 0.0116	9.60×10^{-4}	7.2	Yes
10,000	0 (Uncorr.)	6.01 ± 0.15	21.0848 ± 0.0177	8.79×10^{-3}	2.4	Yes
10,000	1	17.46 ± 0.28	1.6608 ± 0.0781	6.92×10^{-4}	7.0	Yes
10,000	5	19.13 ± 0.15	1.5834 ± 0.0152	6.60×10^{-4}	7.7	Yes
10,000	10	20.26 ± 0.17	1.6134 ± 0.0575	6.72×10^{-4}	8.1	Yes

Table 14: Fixed vs. online-adaptive hyperparameter policies on a LICQ-degraded manifold ($d=500$, $m=200$, true effective rank $r_{\text{eff}}=21$). Adaptive rank $r(t)$ discovers r_{eff} and matches the oracle at machine precision; adaptive damping $\gamma(t)$ accelerates the feasibility transient toward the hand-tuned baseline.

Hyperparameter Policy	Rank r	Proj. Error	$\mathbb{E}[\ h\]$	Subspace W_2^2	Transient $\ h\ $	Steps to Feasible	Stable
Fixed $r=10$ ($< r_{\text{eff}}$)	10.0	1.49×10^{-1}	7.7325	0.3417	—	—	Yes
Fixed $r=21$ ($= r_{\text{eff}}$, Oracle)	21.0	3.86×10^{-15}	0.0000	0.0000	—	—	Yes
Naive Exact G^{-1} ($r=m=200$)	200.0	7.14×10^0	Diverged	Diverged	—	—	No
Adaptive Rank $r(t)$ (Ours)	21.0	6.48×10^{-16}	0.0000	0.0000	—	—	Yes
Fixed Damping $\gamma_0=20$ (Low)	21.0	—	0.0000	0.0000	113.22	94	Yes
Fixed Damping $\gamma_0=100$ (Tuned)	21.0	—	0.0000	0.0000	23.98	17	Yes
Adaptive Damping $\gamma(t)$ ($\gamma_0=20$)	21.0	—	0.0000	0.0000	34.48	22	Yes
Fully Online ($r(t)+\gamma(t)$)	21.0	6.48×10^{-16}	0.0000	0.0000	34.48	22	Yes

F ONLINE ADAPTIVE HYPERPARAMETER SELECTION

This appendix details the online adaptive policies summarized by Ablation 8 and Figure 5. We adapt two hyperparameters without manual tuning. (i) **Adaptive sketch rank $r(t)$** : given a maximum probe budget $r_{\text{probe}} = 40$, we set $r(t) = \#\{\sigma_i(\Omega^T Y) > \tau_{\text{svd}}\}$ with $\tau_{\text{svd}} = 10^{-6}$, so the sketch automatically tracks the effective constraint rank r_{eff} and filters vanishing singular modes. (ii) **Adaptive damping $\gamma(t)$** $= \gamma_0 (1 + c \|h(X_t)\|_2)$: dissipation increases during off-manifold transients and relaxes back to γ_0 near the manifold. Table 14 evaluates fixed versus adaptive policies on a manifold with severe LICQ degradation ($d=500$, $m=200$, 179 redundant equality constraints, $\text{cond}(G) > 10^{18}$, true $r_{\text{eff}}=21$). Adaptive rank recovers the oracle machine-precision projection error while automatically discovering $r_{\text{eff}}=21$, and adaptive damping matches hand-tuned convergence starting from an un-tuned base rate.

Online Adaptive Hyperparameters on a LICQ-Degraded Manifold ($d = 500$, $m = 200$, $r_{\text{eff}} = 21$)

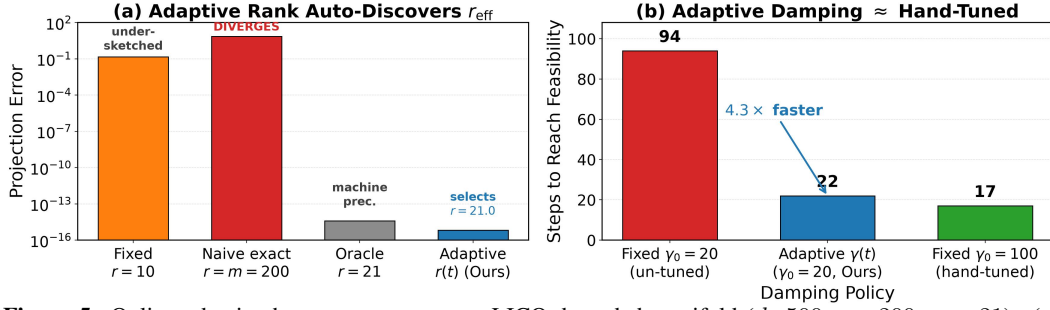


Figure 5: Online adaptive hyperparameters on a LICQ-degraded manifold ($d=500$, $m=200$, $r_{\text{eff}}=21$). (a) Adaptive rank $r(t)$ automatically discovers the effective rank $r_{\text{eff}}=21$ and attains machine-precision projection error, matching the oracle, while fixed under-sketching ($r=10$) drifts and naive exact inversion diverges. (b) Adaptive damping $\gamma(t)$ started from an un-tuned $\gamma_0=20$ reaches feasibility $4.3\times$ faster than fixed $\gamma_0=20$ and nearly matches the hand-tuned $\gamma_0=100$ policy.

High-Dimensional Scaling: Runtime Efficiency and Constraint Precision

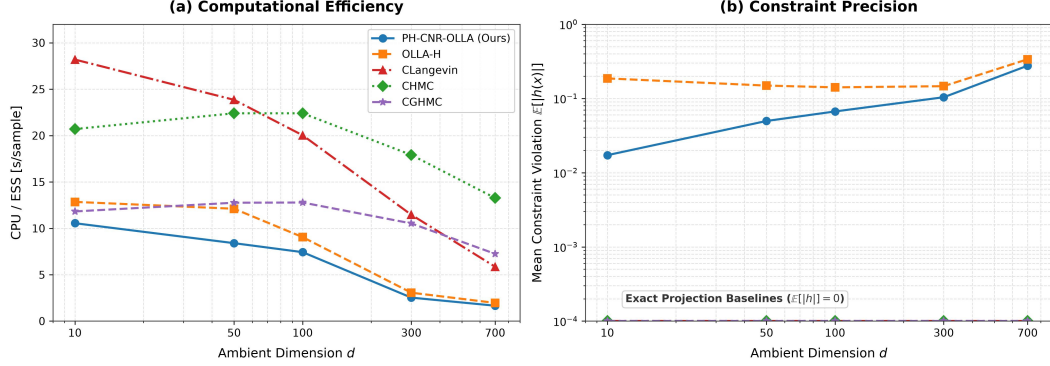


Figure 6: High-dimensional scaling performance on the synthetic stress-test manifold as ambient dimension d increases from 10 to 700. (a) Computational efficiency measured in CPU time per effective sample size (CPU/ESS in seconds per sample), where PH-CNR-OLLA consistently achieves the lowest runtime across all dimensions. (b) Mean equality constraint violation $E[|h(x)|]$, showing that PH-CNR-OLLA maintains tighter constraint precision than OLLA-H, while exact projection baselines (CLangevin, CHMC, CGHMC) remain at zero violation.

G SUPPLEMENTARY FIGURES

This appendix collects the high-dimensional scaling visualization, the qualitative density comparison, and the extended diagnostic studies referenced in Sections 4.2 and 4.3.

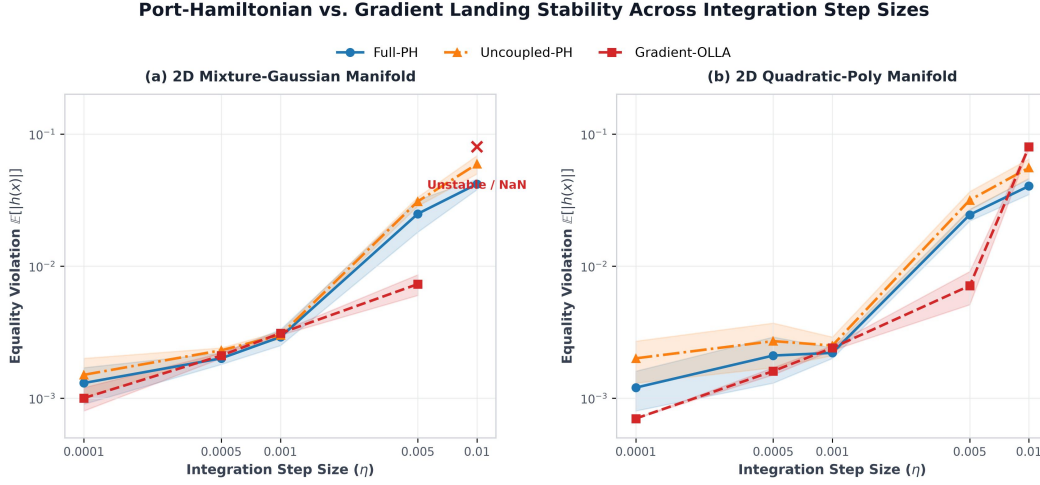


Figure 7: Comparison of landing stability and equality constraint violation across integration step sizes on the 2D Mixture-Gaussian and Quadratic-Poly manifolds. Full Port-Hamiltonian (Full-PH) landing maintains stable execution up to step size $\eta = 0.01$ across both geometries, whereas the Gradient-OLLA baseline suffers from numerical instability and divergence at large step sizes. The skew-symmetric momentum coupling in Full-PH provides tighter equality constraint enforcement compared to the Uncoupled-PH variant, particularly at larger step sizes. Shaded regions indicate plus or minus one standard deviation.

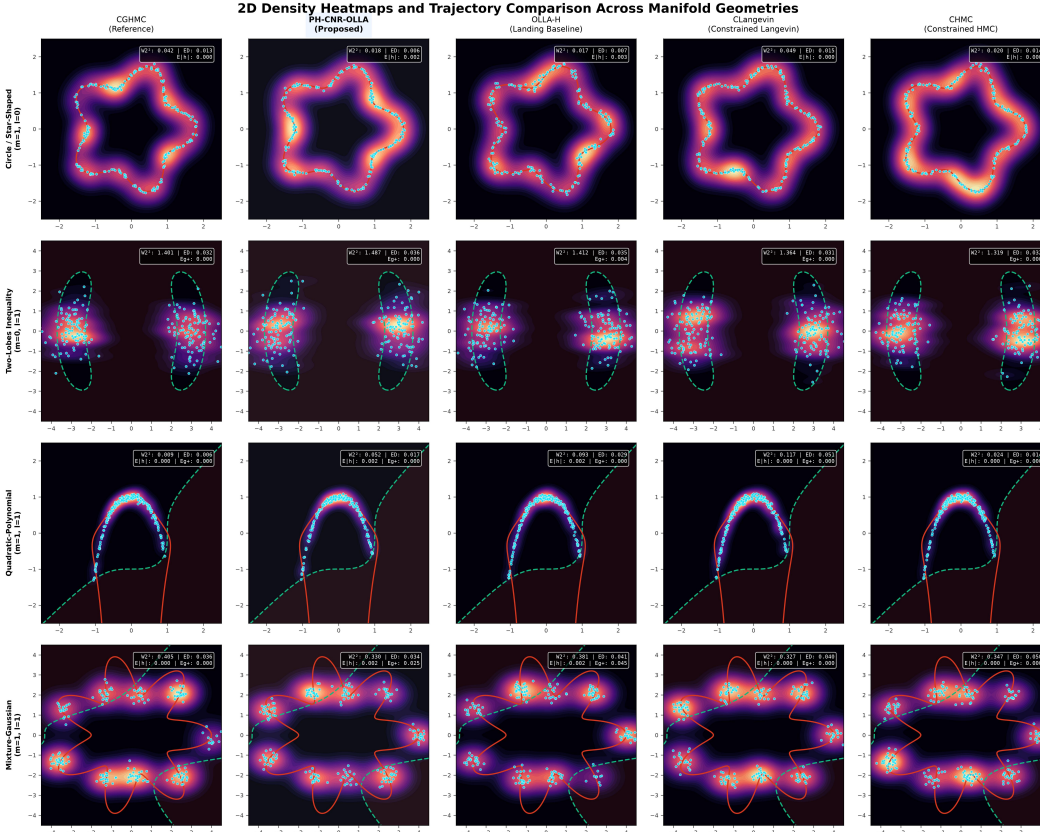


Figure 8: Qualitative comparison of sampled densities (heatmaps) and representative trajectories across the four 2D manifold geometries (rows) for the CGHMC ground-truth reference and the four samplers (columns). Red solid and green dashed curves denote the equality manifold $h(x) = 0$ and the inequality boundary $g(x) = 0$, respectively; cyan points are the retained samples. PH-CNR-OLLA (second column) reproduces the reference density and stays on-manifold across all geometries, matching or improving upon OLLA-H, CLangevin, and CHMC (per-panel W_2^2 , energy distance, and constraint violations are annotated in each panel).

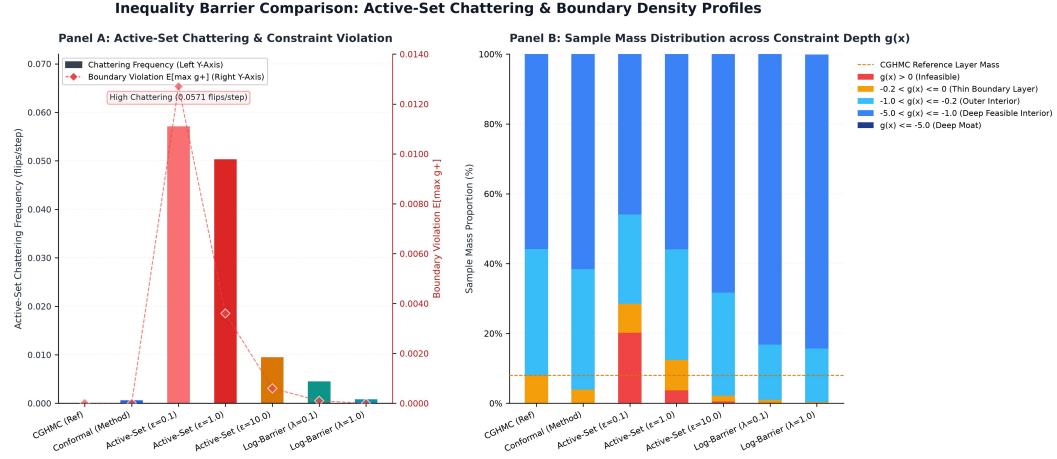


Figure 9: Inequality barrier comparison evaluating active-set chattering frequency, boundary violation, and sample density profiles on the Two-Lobes manifold. Panel A contrasts the active-set indicator chattering frequency in flips per step (left axis, bars) and mean constraint violation $E[\max g^+]$ (right axis, red diamonds) across CGHMC, conformal metric, active-set linear repulsion, and interior log-barrier methods. Panel B shows the stacked distribution of sample mass across constraint depth intervals relative to the ground-truth CGHMC reference. The conformal metric barrier eliminates discrete active-set chattering and prevents boundary penetration while preserving the correct near-boundary sample distribution without interior-penalty or ϵ -offset bias.

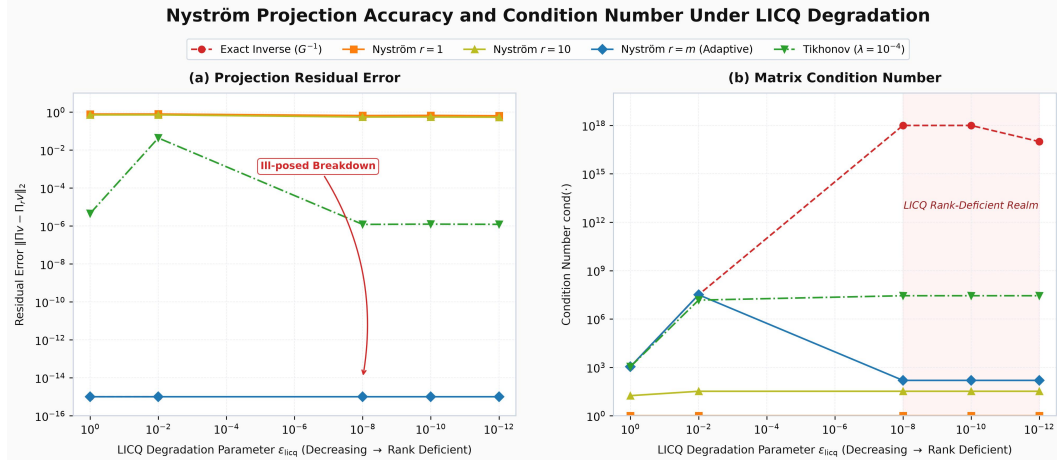


Figure 10: Projection residual error (a) and inverted matrix condition number (b) as linear independence constraint qualification (LICQ) degrades on a 100-dimensional manifold with 60 equality constraints. As constraint gradients become linearly dependent (ϵ_{licq} decreasing to 10^{-12}), exact Gram inversion suffers from extreme matrix ill-conditioning (condition number exceeding 10^{17}), leading to numerical breakdown. In contrast, adaptive Nyström projection ($r = m$) spectrally filters vanishing singular values via rank truncation, bounding the condition number near 1.5×10^2 while maintaining exact tangent projection precision (residual error near machine precision). Fixed low-rank Nyström approximations and Tikhonov regularization prevent numerical explosion under severe rank deficiencies at the expense of higher residual projection error.

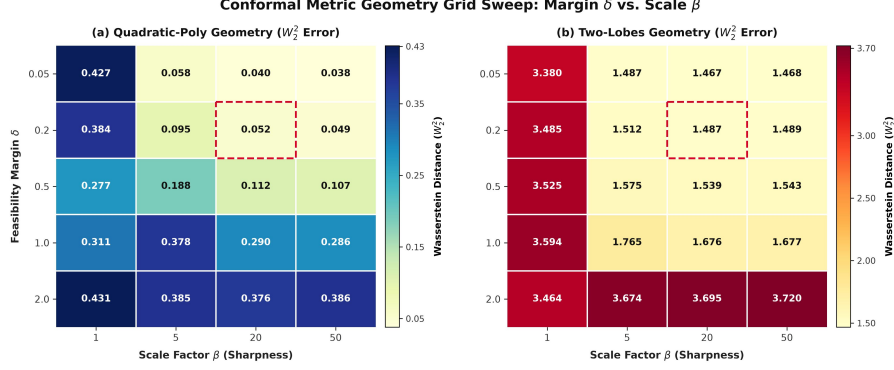


Figure 11: Grid sweep evaluating sampling accuracy (measured by median squared Wasserstein distance relative to the CGHMC ground truth) across feasibility margins δ and scale factors β for the conformal metric barrier on Quadratic-Poly and Two-Lobes 2D geometries. Red dashed boxes highlight the selected operating point ($\delta = 0.2, \beta = 20$), which minimizes distributional error by concentrating boundary repulsion into a thin layer without introducing off-manifold bias.

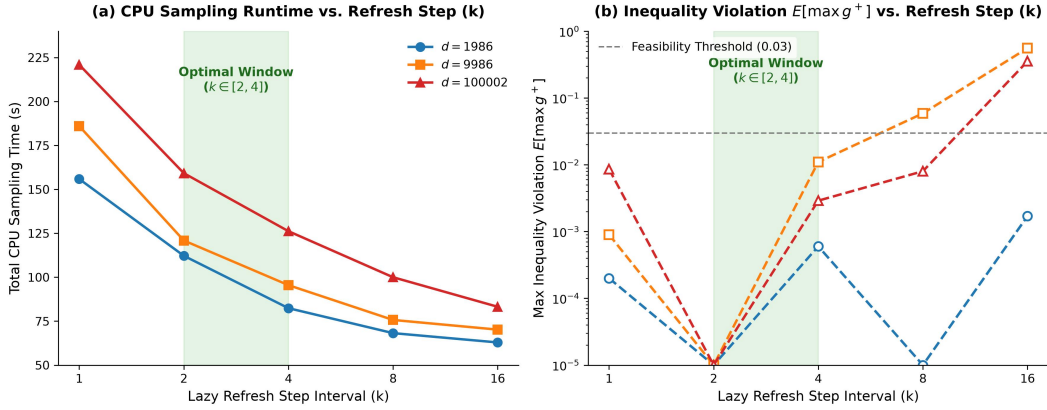


Figure 12: Trade-off between computational efficiency and constraint feasibility under lazy monotonicity-Jacobian refreshing on the German Credit Bayesian logistic regression task across ambient dimensions. (a) Total CPU sampling runtime decreases monotonically as the refresh step interval k increases. (b) Maximum inequality constraint violation $E[\max g^+]$ remains well below the 0.03 feasibility threshold within the highlighted optimal operating window ($k \in [2, 4]$), achieving a 1.9x speedup without compromising constraint satisfaction. Refresh intervals beyond $k = 4$ cause severe constraint violations at higher dimensions.